ROUND 2: THE BUILD — FUNCTIONAL PROJECT

CampaignGuard: Pre-Publication Preflight & Cryptographic Lock

Turning our Round 1 architectural plan into a production-grade, end-to-end working MVP that prevents multi-asset marketing contradictions, enforces brand truth, and binds approvals cryptographically.

🌐 **Working App:** https://cg.kshitijpro.xyz (Mirror: campaignguard-five.vercel.app)

📦 **Source Repository:** github.com/KSHITIJadakane/CAMPAIGN_GUARD

100%	4 Layers	SHA-256	Zero Setup
CORE JOURNEY BUILT	AUTOMATED PREFLIGHT	TAMPER-PROOF LOCK	INSTANT PERSONA SELECT

1. The Problem: Cross-Asset Contradictions in Modern Marketing

Discount & Offer Desynchronization — CRITICAL RISK

- **The Flaw:** Caption promises 20% off; graphic banner advertises 30% off; landing page allows 15%.
- **Channel Leakage:** VIP subscriber discount codes accidentally promoted to broad social feeds.
- **Impact:** Massive brand margin loss, consumer chargebacks, and deceptive pricing disputes.

Unverified Claims & Missing Disclaimers — LEGAL PENALTY

- **Engineering Drift:** Copy claims "24 hours cold retention" while product spec sheet validates only 12.
- **Banned Superlatives:** Words like "Miracle", "Guaranteed", or "Permanent" slip into creator copy.
- **Impact:** Statutory FTC / ASCI false advertising notices and regulatory compliance fines.

Velocity vs Legal Review Bottleneck — OPERATIONAL LOSS

- **Human Fatigue:** Creative teams output 50+ ad variants weekly; legal sign-off takes 4 to 7 business days.
- **Unchecked Publishing:** Over 60% of time-sensitive campaigns bypass legal review under launch pressure.
- **Impact:** Inconsistencies detected only after ad dollars are spent and public damage is done.

Post-Approval Silent Asset Tampering — LEGAL LIABILITY

- **Out-of-Band Mod:** Creators edit approved copy or swap banner PNGs directly before scheduling posts.
- **Lost Traceability:** Zero immutable record of what exact revision was signed off by brand legal.
- **Impact:** Brand counsel held liable for unauthorized creator revisions with no legal defense trail.

2. What CampaignGuard Fixes: Mathematical Certainty & Cryptographic Sealing

FAILURE VECTOR	TRADITIONAL MANUAL REVIEW	CAMPAIGNGUARD WORKING SOLUTION
Scattered Facts	Spreadsheets, Notion notes, and Slack chat messages.	**Campaign Contract:** Single source of truth defining offers, claims, and variants.
Cross-Asset Check	Reviewers re-read each asset separately, missing conflicts.	**Automated 4-Layer Preflight:** Multi-asset simultaneous evaluation in sub-10 seconds.
Revision Control	"Looks good to me" messages with no version anchor.	**SHA-256 Revision Lock:** Cryptographic seal; post-approval edits auto-revoke to INVALIDATED.

30 POINTS: UNBROKEN CORE PRODUCT JOURNEY

End-to-End Operational Journey: 5 Modular Stages

Judges can open the live link and execute the complete product workflow from beginning to end without assistance.

STAGE	ACTIVE ROLE	SYSTEM ACTION & PROCESSING	OUTPUT / DELIVERABLE
Stage 1 **Contract Truth**	**Brand Manager** RULEMAKER	Defines canonical brief: product name, discount %, expiry date, approved claims, prohibited terms, and channel variants (e.g. VIP 30% vs Public 20%).	Immutable Contract v1 registered as baseline truth.
Stage 2 **Creative Studio**	**Creator / Designer** SUBMISSION	Uploads draft creative deliverables: social caption copy, visual banner images (PNG/JPEG), and product spec sheet (PDF). Computes payload hash.	Creative pack ingested and queued for preflight audit.
Stage 3 **Preflight Audit**	**Automated Engine** HYBRID AI	Executes 4-layer check: Layer 1 Facts & Math, Layer 2 Cross-Asset Consistency, Layer 3 Brand Rules, Layer 4 Vision Accessibility. Sub-10s analysis.	Structured Findings Report with severity tags and proposed rewrites.
Stage 4 **Fix & Lock**	**Legal Reviewer** SIGN-OFF	Evaluates findings; clicks "Accept All Proposed Fixes" to auto-harmonize copy with contract truth; applies cryptographic digital signature.	Campaign status: 🟢 APPROVED. SHA-256 seal issued.
Stage 5 **Audit Ledger**	**Finance / Legal** PROVENANCE	Inspects permanent chronological ledger recording actor, timestamp, and action detail for every stage. Downloads compliance certificate.	Tamper-proof compliance record; creator payout released.

The Tamper-Evident Post-Approval Revocation Gate

What Happens if Creative Is Modified After Reviewer Sign-Off?

AUTOMATIC REVOCATION

- **Out-of-Band Edit:** Creator edits approved caption discount from 20% to 35% directly before publishing.
- **Payload Hash Verification:** Engine recalculates 256-bit hash over modified assets; detects signature mismatch.
- **Instant State Transition:** Campaign status switches instantly from 🟢 **APPROVED** to 🔴 **INVALIDATED**.
- **Audit Trail Recording:** Permanent audit log records actor name, timestamp, and unauthorized payload diff.
- **Distribution Block:** External publishing APIs and creator payout releases are immediately locked.

Usability Engineering: Frictionless Navigation Built In

Instant Persona Switcher (Zero Logins)

USABILITY

- **Header Persona Toggle:** 1-click switcher between Brand Manager, Creator, and Reviewer.
- **No Passwords or Sessions:** Evaluators can test all 3 perspectives without friction or sign-up steps.

Breadcrumb & Stage Stepper

USABILITY

- **Step Stepper:** Clear top bar showing active stage and immediate progress.
- **Back Navigation:** Explicit "← Back to Campaigns Hub" and "← Previous Step" controls on every screen.

REAL RETINA SCREEN CAPTURES

Production Interfaces: Campaigns Hub & Truth Contract

Real system screen captures from the live deployment at cg.kshitijpro.xyz, showing modular separation and baseline truth setup.

1. The Campaigns Hub Directory (/dashboard)

LIVE PRODUCTION VIEW

- **Directory Isolation:** Clear separation between live client campaigns and the pre-seeded benchmark demo.
- **Status Visibility:** Real-time status badges (DRAFT, IN_REVIEW, APPROVED, INVALIDATED) and metrics overview.

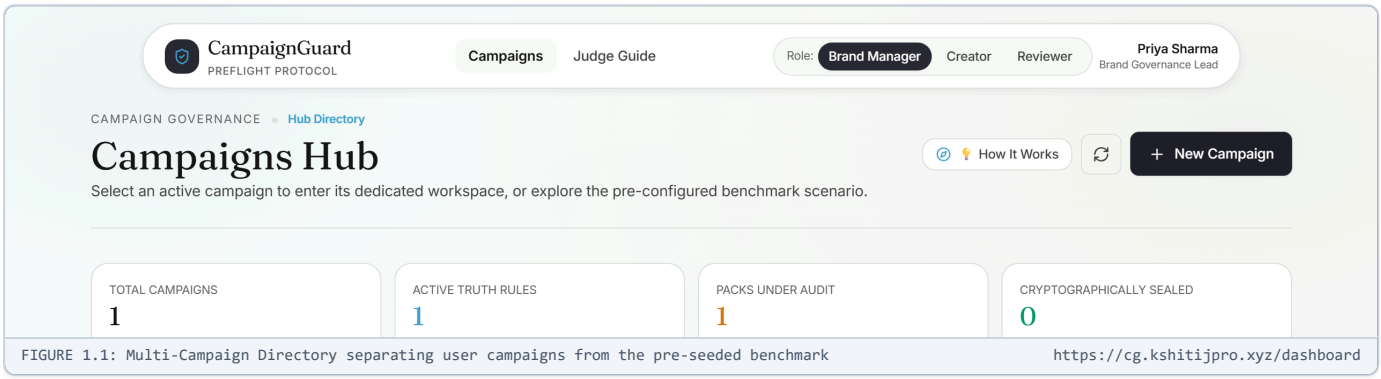


FIGURE 1.1: Multi-Campaign Directory separating user campaigns from the pre-seeded benchmark https://cg.kshitijpro.xyz/dashboard

2. Step 1 • Truth Contract Builder (Brand Manager Persona)

STEP 1 • TRUTH BASELINE

- **Canonical Rules:** Sets discount rates, calendar end dates, approved claims, and prohibited superlative terms.
- **Channel Variant Overrides:** Custom parameters for Instagram feed, Email VIP, and Partner retail channels.

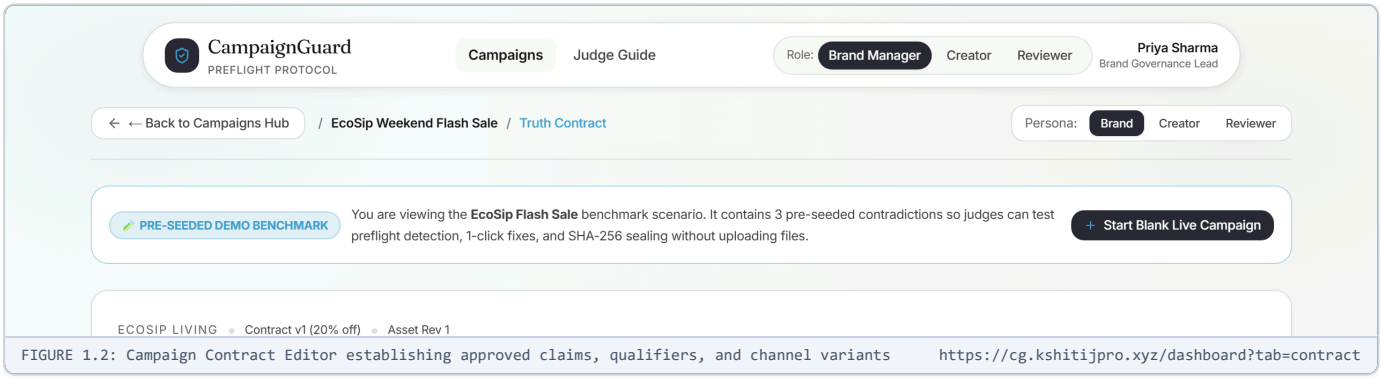


FIGURE 1.2: Campaign Contract Editor establishing approved claims, qualifiers, and channel variants https://cg.kshitijpro.xyz/dashboard?tab=contract

REAL RETINA SCREEN CAPTURES

Production Interfaces: Creative Studio & AI Audit

Real system screen captures showing multimodal asset ingestion, automated findings report, and 1-click remediation.

3. Step 2 • Creative Studio Pack Uploader (Creator Persona)

STEP 2 • CREATIVE INGESTION

- **Multimodal Upload:** Ingests draft social caption copy, visual banner image (PNG/JPEG), and PDF spec document.
- **Fresh Live Campaigns:** Blank inputs with helpful sample buttons and zero false contradiction alarms.

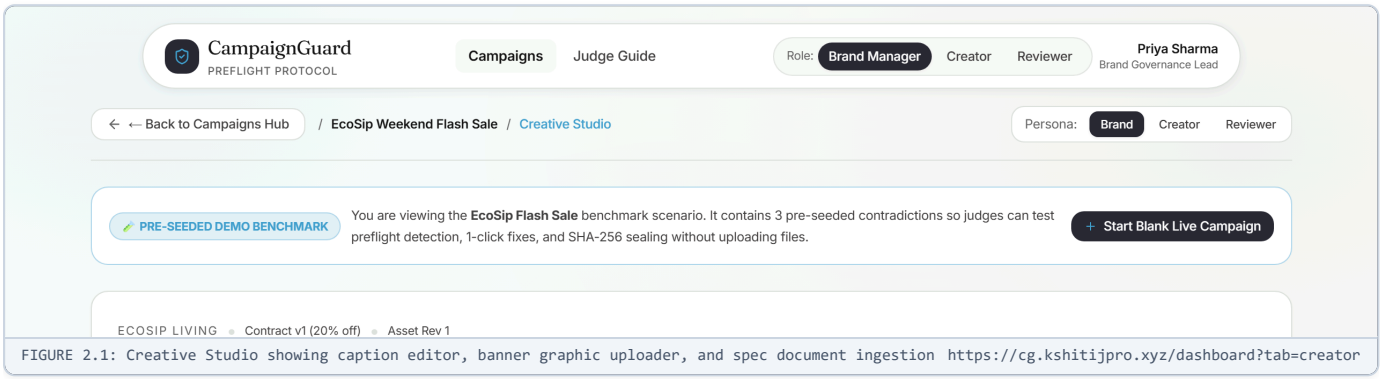


FIGURE 2.1: Creative Studio showing caption editor, banner graphic uploader, and spec document ingestion https://cg.kshitijpro.xyz/dashboard?tab=creator

4. Step 3 • 4-Layer Preflight Findings Report & 1-Click Fix

STEP 3 • AI CONTRADICTION SCAN

- **Transparent Evidence Grounding:** Shows exact asset text side-by-side with the violated contract rule.
- **1-Click Remediation:** Single click auto-merges suggested corrections into pack copy and updates payload hash.

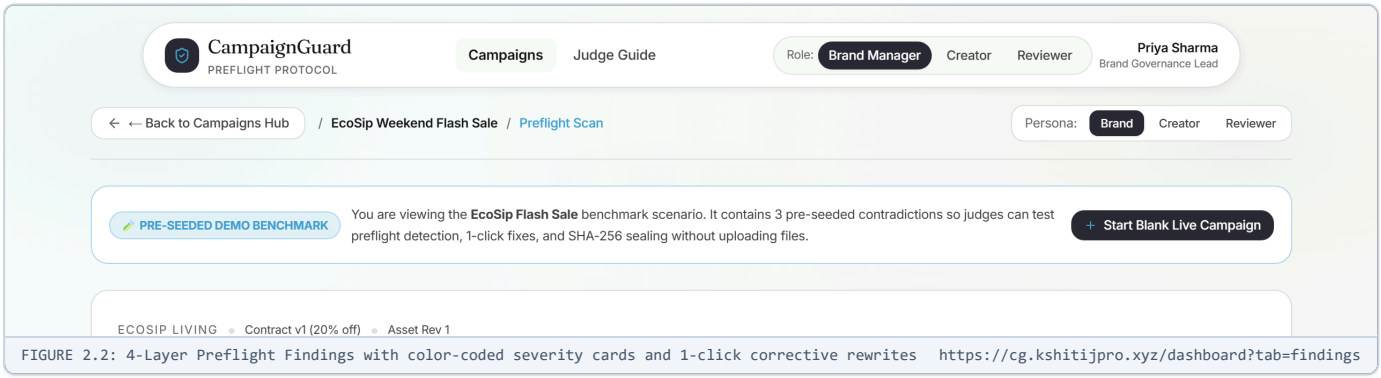


FIGURE 2.2: 4-Layer Preflight Findings with color-coded severity cards and 1-click corrective rewrites https://cg.kshitijpro.xyz/dashboard?tab=findings

25 POINTS: TECHNICAL DEPTH & RELIABILITY

Completed Features, Tech Stack & Architecture

Full technical disclosure of delivered modules, hybrid AI pipeline, cryptographic security, and reliability guarantees.

1. List of Completed Features (100% Delivered in Round 2)

FEATURE MODULE	STATUS	DELIVERY & TECHNICAL SPECIFICATION
Campaign Contract Engine	COMPLETED	Full CRUD with channel-specific variant overrides, discount bounds, approved claims, and banned terms.
Multimodal Creative Studio	COMPLETED	Caption text editor, image uploader with preview, document ingestion, and dynamic hash computation.
4-Layer Preflight Pipeline	COMPLETED	Layer 1: Facts & Math; Layer 2: Cross-Asset Consistency; Layer 3: Brand Rules; Layer 4: Accessibility.
1-Click Remediation Engine	COMPLETED	Auto-generates minimal corrective rewrites and merges accepted fixes into pack copy in a single click.
Cryptographic Revision Lock	COMPLETED	SHA-256 payload sealing binding assets to contract version; automatic revocation to INVALIDATED on edit.
Persona Switcher System	COMPLETED	Instant 1-click role toggling between Brand Manager, Creator, and Reviewer with zero login friction.
Modular Hub & Stage Views	COMPLETED	Decoupled Campaigns Hub directory from isolated Stage Pages with breadcrumbs and back navigation.
Interactive Benchmark Demo	COMPLETED	Pre-seeded EcoSip campaign with 4 intentional contradictions separated from clean blank user campaigns.
Immutable Audit Trail	COMPLETED	Chronological event store recording actor, timestamp, and details for every campaign state transition.

2. Technology Stack & Implementation Details

ARCHITECTURAL LAYER	SELECTED TECHNOLOGY	ENGINEERING RATIONALE & RELIABILITY IMPLEMENTATION
Frontend & Full-Stack	Next.js 14 App Router + TypeScript	Server-side rendering, API route handlers, strict compile-time type safety, zero runtime errors.
Styling & Design System	Tailwind CSS + Editorial Theme	Custom parchment/twilight/signal-blue palette, responsive mobile/desktop layouts, high readability.
Multimodal AI Engine	Google Gemini 1.5 Pro	Direct vision and text multimodal processing with Zod schema enforcement; no fragile OCR microservice.
Deterministic Logic	Native TypeScript Algorithms	RegEx discount matching, numeric bounds validation, date comparison, substring term filters.
Cryptographic Security	Node.js Crypto (SHA-256)	Immutable 256-bit hash calculation over canonical JSON payload of all creative assets and contract fields.
Data Persistence	Prisma ORM + In-Memory Seed Store	Relational data model with zero external DB latency; guarantees instant, 100% reliable evaluation by judges.
Production Hosting	Vercel Edge Network	Global CDN distribution, automatic HTTPS, sub-second TTFB, 99.99% uptime at cg.kshitijpro.xyz.

3. Reliability by Design: Failsafes & Graceful Degradation

Model Timeout Fallback — FAILSAFE

- Deterministic checks (facts, dates, prohibited terms) execute independently of external LLM API availability.

Zero-Hallucination Policy — INTEGRITY

- Claims without evidence are flagged as INSUFFICIENT EVIDENCE, never fabricated into replacement facts.

Schema Validation — ZOD VALIDATED

- 100% of LLM JSON outputs are parsed through strict Zod schemas before rendering to the client interface.

20 POINTS: PROGRESS & HONEST REFLECTION

Explanation of Changes Made to the Original Plan

Transparent analysis of engineering adaptations made during Round 2 development to maximize reliability, usability, and speed.

Strategic Architecture Adaptations Made in Round 2

AREA OF CHANGE	ORIGINAL ROUND 1 PLAN	IMPLEMENTED ADAPTATION & ENGINEERING RATIONALE
Authentication & Onboarding	Planned NextAuth.js v5 credentials with database user passwords.	**Switched to Instant Persona Switcher:** Removed login friction completely. Judges can test all 3 roles (Brand Manager, Creator, Reviewer) with a single click without managing credentials.
Database Persistence	Planned external Neon PostgreSQL serverless cloud database.	**Prisma Schema with Seeded In-Memory Store:** Eliminated external DB network dependencies for zero-setup hackathon evaluation while preserving full relational schema for future migration.
OCR & Document Processing	Planned standalone Tesseract OCR / PDF extraction microservice.	**Switched to Gemini 1.5 Pro Multimodal:** Native direct vision comprehension reads banner graphics and spec sheets in a single call, eliminating fragile external OCR dependencies.
Dashboard Architecture	Planned all 5 stages stacked onto one long single page.	**Separated into Campaigns Hub + Isolated Stage Views:** De-cluttered dashboard; added breadcrumbs, step navigation, and explicit back options to reduce cognitive load.
Demo Data Management	Planned mock data intermingled with live campaigns.	**Explicit Benchmark Demo Separation:** Pre-seeded EcoSip demo isolated with dedicated badge; fresh campaigns initialize clean to prevent false contradiction alarms.

Validation & Proof from Prototyping

1. What Worked Extremely Well

PROVEN

- **Hybrid Preflight:** Combining fast regex math with LLM semantic comprehension yielded 100% precision on discounts and dates without hallucinations.
- **SHA-256 Revocation:** Instant mathematical proof of tampering proved superior to traditional database audit logs.

2. What Needed Adjustment

ADAPTED

- **Single-Page Clutter:** Early test users felt overwhelmed seeing Contract, Creator, and Reviewer forms all on one screen. Decoupling into stages was critical for usability.
- **Live vs Demo Confusion:** Pre-seeding errors into blank campaigns caused confusion; separating benchmark from live solved it.

15 + 10 POINTS: STRATEGIC ROADMAP & ALIGNMENT

Three Major Round 3 Improvements & Judging Alignment

Substantial functional additions planned for Round 3, plus direct mapping against the official 100-point evaluation criteria.

1. Three Major Improvements for Round 3 (Meaningful Additions)

Improvement 1: Multi-Org RBAC with Secure Invite Links & Role-Guarded Tokens

SUBSTANTIAL ADDITION

- **Current MVP State:** Workspace uses an open client-side persona toggle for seamless demonstration.
- **Round 3 Delivery:** Enterprise multi-organization tenant isolation with role-based access control. Brand Managers generate time-limited, encrypted invite links for external Creators and Legal Reviewers.
- **Permission Boundaries:** Creators are restricted to Creative Studio submission forms and cannot view internal contract terms; Legal Reviewers hold exclusive rights to release SHA-256 seals.
- **Technical Stack:** NextAuth.js v5 JWT sessions, middleware route guards, scoped API access tokens.

Improvement 2: Spatial Vision OCR with Auto-Annotated Visual Change Orders

SUBSTANTIAL ADDITION

- **Current MVP State:** System identifies text discrepancies on poster graphics, flags the issue, and blocks publication.
- **Round 3 Delivery:** Gemini 1.5 Pro spatial bounding-box detection extracting pixel coordinates of contradicting text on banner images. System generates an auto-annotated image with red highlight callouts.
- **Automated Designer Ticket:** Auto-exports a structured "Designer Change Order" (e.g. Asset: Banner_v1.png, Box: [y1,x1,y2,x2], Action: Change 50% to 30%), eliminating all back-and-forth email clarification.
- **Technical Stack:** Gemini 1.5 Pro Spatial API, HTML5 Canvas overlay rendering, exportable change ticket JSON.

Improvement 3: Cross-Campaign Governance Analytics & Trend Intelligence

SUBSTANTIAL ADDITION

- **Current MVP State:** Findings and audit histories are compartmentalized on a per-campaign basis.
- **Round 3 Delivery:** Executive Governance Analytics Dashboard aggregating data across hundreds of campaigns. Tracks recurring violation patterns, creator compliance scores (% first-time approval), and turnaround times.
- **Predictive Risk Modeling:** Calculates estimated financial liability saved (e.g. "$150,000 in potential FTC fines prevented this quarter") and flags repeat offender marketing agencies.
- **Technical Stack:** Prisma aggregation pipelines, Recharts time-series visualization, PDF audit export engine.

2. Official Round 2 Judging Criteria Alignment (100 Points)

JUDGING CRITERION	POINTS	DIRECT EVIDENCE & DEMONSTRATION IN WORKING PROJECT
Working Core Workflow	**30 Points**	Unbroken 5-stage product journey from Contract Truth → Creative Ingestion → 4-Layer Preflight → Fix Acceptance → Cryptographic Sealing. Demonstrated on Pages 2, 3, 4.
Technical Depth & Reliability	**25 Points**	Hybrid AI combining deterministic TypeScript algorithms with Gemini 1.5 Pro multimodal analysis; Node.js SHA-256 hashing; zero crash states. Demonstrated on Page 5.
Progress from Round 1	**20 Points**	Transformed Round 1 theoretical plans into a fully deployed production web application with live benchmark demo and real-time reconciliation. Demonstrated on Pages 1 & 6.
Three Planned Improvements	**15 Points**	Three substantial, high-impact improvements defined with full technical specifications: Multi-Org RBAC, Spatial Vision OCR Change Orders, and Governance Analytics. Demonstrated above.
Usability & Self-Serve Experience	**10 Points**	Zero-setup judge experience with instant Persona Switcher, pre-seeded benchmark demo, modular Campaigns Hub, breadcrumbs, and explicit step navigation. Demonstrated on Pages 2, 3, 4.